



# NYC Public Computer Centers

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Capstone Project

# Introduction

As a Computer Information Systems student, my goal is to give back to my community by sharing my knowledge and contributing to public-serving institutions. So far, this has been through the Free Library of Philadelphia and related industries that provide free, accessible services to the public.

Public computer centers offer a variety of free technology related courses and services that bridge the digital divide, including:

- Productivity Tools (ex: Using Word, Excel, PowerPoint, Adobe Acrobat, etc.),
- Job Readiness (ex: resume help, job search, etc.),
- Education (ex: personal growth, k-12 supports, reading research, etc.)',
- Certifications (ex: in software, in housing, in professional areas, etc.)',
- Digital Literacy Courses

Community centers that offer these services are vital to their neighborhoods, and they should be seen as models for public access and equity in digital education. In this project, we examine public computer centers across New York City, focusing on the range of resources they offer, the average number of services available per borough, and top performers throughout the city.

# Data sources

Using NYC Open Data, a free public data published by New York City agencies, we can gather data on public computer centers that offer tech resources.

[https://data.cityofnewyork.us/Social-Services/Citywide-Public-Computer-Centers/sejx-2gn3/about\\_data](https://data.cityofnewyork.us/Social-Services/Citywide-Public-Computer-Centers/sejx-2gn3/about_data)

The Citywide Public Computer Centers database lists facilities offering computers with internet access, according to administrative data collected by the Office of Technology and Innovation.



# Methodologies

Using pandas and matplotlib, we can manipulate and visualize the data in the Citywide Public Computer Centers database. After cleaning the data, we can identify each resource provided by each center and tally them.

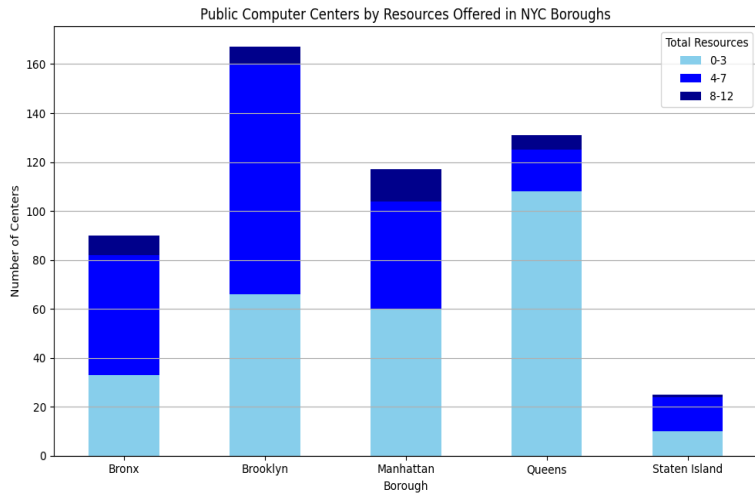
Depending on the number of resources provided, each center is grouped into one of three categories (0-3, 4-7, or 8-12, with 12 representing the full range of resources listed by the CPCC database).

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
#cleaning
dfResources = dfResources.fillna('No')
dfResources = dfResources.replace(
    to_replace=r'(?i)^Not Sure$', value='No', regex=True
)
```

```
#number of resources categories
def categories(x):
    if x <= 3:
        return '0-3'
    elif x <= 7:
        return '4-7'
    else:
        return '8-12'

df['numResources'] = df['Total Resources'].apply(categories)
```



```
#visualization
resourceCounts.plot(
    kind='bar',
    stacked=True,
    figsize=(10, 6),
    color=['skyblue', 'blue', 'darkblue']
)

plt.title('Public Computer Centers by Resources Offered in NYC Boroughs')
plt.xlabel('Borough')
plt.ylabel('Number of Centers')
plt.legend(title='Total Resources')
plt.xticks(rotation=0)
plt.tight_layout()
plt.grid(axis='y')
plt.show()
```

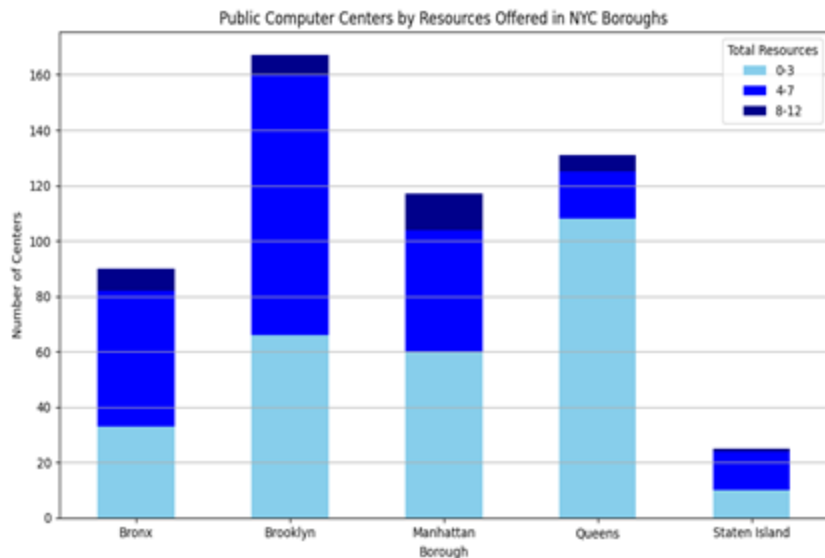
## Methodologies Pt. 2

These groupings allow us to generate a stacked bar chart that illustrates the distribution of centers by borough and the extent of services offered at each. This visualization gives insights into both access to resources and geographic disparities across NYC.

# Results

The graph below and the accompanying averages differentiate in that the averages don't take the number of centers into account. Visualizing the data through these methods was necessary to get the full picture.

Brooklyn has the most computer centers, and averages ~4 tech resources per center. Brooklyn's Tompkins Park Neighborhood Senior Center has the most listed resources for its community, which is 10 programs.



```
print(df.groupby('Borough')['Total Resources'].mean().sort_values(ascending=False))
```

```
Borough
Bronx      4.222222
Brooklyn   4.047904
Manhattan  3.991453
Staten Island 3.880000
Queens     2.648855
Name: Total Resources, dtype: float64
```

```
topLocations = df.sort_values(by='Total Resources', ascending=False)
topLocations[['Location Name', 'Borough', 'Total Resources']].head()
```

|     | Location Name                            | Borough   | Total Resources |
|-----|------------------------------------------|-----------|-----------------|
| 66  | TOMPKINS PARK NEIGHBORHOOD SENIOR CENTER | Brooklyn  | 10              |
| 491 | CHELSEA                                  | Manhattan | 9               |
| 494 | STARVOS NIARCHOS FOUNDATION LIBRARY      | Manhattan | 9               |
| 461 | WOODSIDE NEIGHBORHOOD SENIOR CENTER      | Queens    | 9               |
| 497 | ANDREW HEISKELL                          | Manhattan | 9               |

# To Conclude

New York City has a vast network of libraries, community centers, and organizations that offer free access to computers and digital services.

While many of these public computer centers offer a limited range of support, often fewer than three of the twelve key resources identified in the CPCC database, certain areas demonstrate a stronger commitment to digital inclusion. Notably, the borough with the largest population, Brooklyn, leads in offering the most comprehensive range of technology-related programs. This suggests a correlation between population density and investment in digital resources. Larger communities are investing more in public computer access and may receive more support in terms of education, job readiness, and digital literacy.

These resources are vital to the communities they serve. By providing access to technology and educational tools, they empower individuals to pursue self-guided learning, enhance their skills, and invest in their personal and professional development. Access to such opportunities should be equitable and available to all.

Thank you for your time and attention.